

# Linear Stability Analysis Solver for Cylindrical Liquid Jets

## Overview

This repository contains a MATLAB implementation of a **Linear Stability Analysis (LSA)** solver for cylindrical liquid jets using **Muller's root-finding method**. The code computes the dispersion relation, identifies unstable modes, determines the dominant instability characteristics, and predicts key atomization parameters such as:

- Maximum growth rate
- Dominant wavelength
- Disturbance frequency
- Wave propagation velocity
- Breakup length

The solver was developed for studying the primary breakup of liquid jets and can be extended to non-Newtonian fluids, coaxial gas-assisted atomizers, and other multiphase flow configurations.

---

## Features

- Solves the characteristic dispersion equation using **Muller's Method**
  - Uses continuation-based root tracking for improved convergence
  - Automatically identifies unstable modes
  - Classifies disturbances into:
    - Para-sinusoidal mode
    - Para-varicose mode
  - Computes dimensional instability characteristics
  - Generates dispersion curves
  - Exports results directly to Excel
- 

## Governing Framework

The code solves the linearized dispersion relation obtained from the stability analysis of a cylindrical liquid jet.

For each non-dimensional wavenumber:

1. The characteristic equation is assembled.
2. Muller's method is used to locate the complex root:

$$S = S_r + iS_i$$

where:

- $S_r$  = temporal growth rate
- $S_i$  = oscillation frequency
- Positive values of  $S_r$  correspond to unstable disturbances.

The dominant instability is identified as the mode with the maximum growth rate.

---

## Required Files

The main script depends on several auxiliary Bessel-function wrapper routines.

Example structure:

```
Repository/
|
├─ DispEq_Mullers.m
|
├─ I0Lh.m
├─ I1Lh.m
├─ I0L.m
├─ I1L.m
├─ K0Lh.m
├─ K1Lh.m
├─ K0L.m
├─ K1L.m
├─ I0Mh.m
├─ I1Mh.m
├─ K0k.m
├─ K1k.m
|
└─ README.md
```

Ensure all helper functions are available in the MATLAB path before running the solver.

---

## Input Parameters

The physical properties and operating conditions are specified in the first section of the script.

## Fluid Properties

rho\_l  
sigma  
K\_const  
n

| Parameter | Description                         |
|-----------|-------------------------------------|
| rho_l     | Liquid density (kg/m <sup>3</sup> ) |
| sigma     | Surface tension (N/m)               |
| K_const   | Consistency index                   |
| n         | Flow behavior index                 |

## Jet Geometry

d  
b  
h

| Parameter | Description          |
|-----------|----------------------|
| d         | Orifice diameter (m) |
| b         | Jet radius (m)       |
| h         | Radius ratio         |

## Flow Conditions

U\_l  
Qi  
Qo  
We\_l  
Re\_n

These quantities define the operating point of the atomizer or jet.

## Running the Solver

Simply execute:

DispEq\_Mullers

The code will:

1. Sweep the specified wavenumber range.
2. Solve the dispersion relation.
3. Identify unstable roots.
4. Plot the dispersion curve.
5. Compute dominant instability parameters.
6. Prompt the user to save results to Excel.

---

## Outputs

### Dispersion Curve

The solver plots:

Growth Rate vs Wavenumber

showing the unstable region and dominant mode.

---

### Instability Parameters

For the maximum growth-rate mode, the code calculates:

#### Growth Rate

$$S_{r,max}$$

#### Wavelength

$$\lambda = \frac{2\pi}{k}$$

#### Disturbance Frequency

$$f = \frac{\omega_i}{2\pi}$$

#### Wave Propagation Velocity

$$V_p = \frac{\omega_i}{k}$$

## Breakup Length

$$BL = \frac{12U_l}{S_{r,max}}$$

---

## Excel Export

The program automatically exports:

### Sheet 1: Data

| Column     | Description                 |
|------------|-----------------------------|
| K_vec      | Non-dimensional wavenumber  |
| S_sin_real | Sinusoidal growth rate      |
| S_sin_imag | Sinusoidal frequency        |
| ...        | Additional mode information |

### Sheet 2: Summary

| Parameter  | Description               |
|------------|---------------------------|
| BL         | Breakup length            |
| lambda_max | Dominant wavelength       |
| f_dist_max | Disturbance frequency     |
| Vp_max     | Wave propagation velocity |

---

## Numerical Method

### Muller's Method

The dispersion relation is complex and highly nonlinear.

Muller's method is employed because it:

- Handles complex roots directly
- Does not require derivative evaluation
- Provides robust convergence for stability problems

A continuation strategy is implemented whereby the converged root at a given wavenumber is used as the initial guess for the next wavenumber.

This significantly improves convergence and mode tracking.

---

## Applications

This solver can be used for:

- Pressure-swirl atomizers
  - Air-blast atomizers
  - Coaxial jets
  - Liquid sheet stability studies
  - Non-Newtonian jet breakup
  - Primary atomization research
  - Validation of instability-based breakup models
- 

## Citation

If you use this code in academic work, please cite the associated thesis, journal article, or publication from which the dispersion relation and model formulation were developed.

---

## Author

Edin Michael

Department of Mechanical and Aerospace Engineering

Indian Institute of Technology Hyderabad

---

## License

This project is intended for academic and research use. Please contact the author for permissions regarding commercial use or redistribution.